OP0P CC-CLUSTER

Tune your radio with one click

A simple step-by-step guide, for everyone - click a frequency on the cluster (from your PC or your phone) and your transceiver tunes to it automatically.

cluster.op0p.be

What is this, and what is it for?

> The OP0P cluster (**cluster.op0p.be**) shows frequencies spotted live by other radio amateurs. Normally, to listen to a station that was spotted, you have to tune your radio to that frequency yourself. With the small program described in this guide, a single click on the frequency (from your computer, or even from your phone) tunes your radio there automatically.

What you need before you start:

- A **Windows** computer, connected to your transceiver with the usual cable (USB/CAT).
- Your radio already switched on and recognized normally by your computer (as usual for controlling it from software).
- An internet connection (just long enough to download the program, once).

Installation (one time only)

1 Install Hamlib

This is the software that knows how to talk to virtually every radio brand (Icom, Yaesu, Kenwood, Elecraft, ...). Download it from github.com/Hamlib/Hamlib/releases (pick the Windows file, w64, the most recent one) and install it like any other program (double-click, "Next" through the wizard).

2 Download the CAT bridge program

Go to cluster.op0p.be/releases and click **"Download PontCAT-OP0P.exe"** (the first option, for Windows - no other installation is needed).

3 Launch the program

Double-click the downloaded file (`PontCAT-OP0P.exe`, in your Downloads folder). A small window opens.

4 Choose your radio

In the window: select the **brand** and then the exact **model** of your radio from the two drop-down lists, and check the **port** (COM1, COM3, ...) that is already suggested - it normally matches your cable.

5 Start

Click the **Start** button. After a few seconds, your browser opens on its own on the cluster page: you're ready. Click a frequency in the spot list - your radio should tune to it right away.

> Keep this window open for as long as you want to use the cluster with your radio. To stop, click "Stop" or simply close the window.

Also controlling it from your phone optional

6 Allow access from your WiFi network

Before clicking Start (step 5), tick the box **"Accessible from the local network (phone, tablet...)"** and enter a short code of your choosing (for example `1234`) in the **PIN code** box.

7 Open the address on your phone

Once started, a window appears showing an address such as `http://192.168.1.XX:5005/?pin=1234`. On your phone, connected to the **same WiFi network** as this computer (this is required), open that address in your browser, **once**.

That's it: the cluster page appears on your phone, and you won't need to type the code again on that device afterwards. Clicking a frequency from the phone tunes the radio, exactly as from the PC.

If something goes wrong

What you see	What to do
Message: the program `rigctld` could not be found	Hamlib is not installed (or not found) - redo step 1.
No port detected / radio not responding	Check that the USB cable is properly plugged in and your radio is switched on; restart the program.
Your phone shows "PIN required"	Add `?pin=your_code` to the end of the address in the phone's browser, once.
Nothing happens when clicking a frequency	Make sure the program's window is still open on the computer connected to the radio.

Guide for **cluster.op0p.be** (OP0P CC-Cluster). For any question, contact ON6DP / OP0P. Full setup guide online: cluster.op0p.be/cat.html.